SAHIL KUMAR

MCA CANDIDATE (2027) | SOFTWARE & IT

Bengaluru, India | +91 7321843138 | sahilsea10@gmail.com
linkedin.com/in/sahil-kumar-9a10e4 | github.com/sahilsea

PROFILE

MCA candidate graduating in 2027, seeking entry-level roles in software development and IT. Project experience in Python and JavaScript, modular REST APIs, AI applications and automated testing, supported by foundations in OOP, DSA, SDLC and Agile/Scrum. Built secure AI workflows, tested core modules and optimized database performance. Experience in app testing, website maintenance and collaboration with a US team; eager to learn new technologies.

EDUCATION

Master of Computer Applications (MCA) 2025 - 2027 (Expected)
Presidency University, Bengaluru | **CGPA: 7.91**

Bachelor of Computer Applications (BCA) 2022 - 2025
Aryabhatta Knowledge University, Patna | **CGPA: 8.01**

TECHNICAL SKILLS

- **Programming & APIs:** Python, SQL, JavaScript, Java, C++; Flask, FastAPI, Node.js, Express.js, REST APIs
- **AI & Core Concepts:** Google ADK, LLMs, RAG, Pandas, NumPy, Scikit-learn; OOP, DSA, SDLC, Agile/Scrum
- **Automation & Troubleshooting:** Python, Pytest, unit testing, debugging, fault recovery
- **Databases & Tools:** PostgreSQL, MySQL, SQLite, MongoDB, Git, GitHub
- **Infrastructure Foundations:** Networking fundamentals, Docker, Docker Compose, CI/CD pipelines
- **Security:** Access control, input validation, SHA-256 audit logging, secure document retrieval

RELEVANT CERTIFICATIONS

LinkedIn Learning | July 2024: Networking Foundations: Networking Basics; Learning Docker; Learning Docker Compose; Docker: Your First Project; Web3 Design and Security Essentials.

PROJECTS

SEVERANCE - Air-Gapped Document Trust Workbench | GitHub 2026
Smart India Hackathon 2026 (MRPL Problem Statement) | Python, FastAPI, Google ADK, SQLite, Pydantic, Pytest

- Developed Python access-control modules to keep unauthorized documents out of retrieval results and LLM context.
- Built FastAPI / Google ADK backend validation with Pydantic, exact-substring citation checks and deterministic retries to reject unverified AI output.
- Built a SHA-256 hash-chained SQLite audit ledger; wrote 60+ Pytest cases for security gates and fault recovery, achieving 85% coverage on core modules.

Credit Card Fraud Detection System 2024
Python, Flask, REST APIs, Pandas, NumPy, Scikit-learn, Deep Learning

- Developed a Flask REST API for real-time fraud scoring with response times under 200 ms; trained and evaluated a model achieving 95% recall on a held-out test set.
- Prepared and balanced transaction data; engineered features for frequency, amount spikes, new devices and impossible travel to support fraud detection.

Full-Stack Web Application (MERN) 2025
Node.js, Express.js, MongoDB, REST APIs, React.js, Git

- Designed 12 modular REST endpoints using Node.js and Express.js in a layered architecture; indexed MongoDB schemas to reduce average query latency by 40% on frequent read paths.

EXPERIENCE

Product & Digital Media Associate (Remote) | TRACE, United States Mar 2025 - Sep 2025

- Maintained company website content across product releases, keeping product information current for users.
- Tested the TRACE app and proposed feature and UX improvements to founders; collaborated remotely with a US team across time zones to share product feedback.
- Created 30+ video ads and product demos over six months, translating product documentation and features into clear explanations for end users.